

TURKU RAJAKARI, Finland

WMO: 027470

Lat: 60.378N

Lon: 22.096E

Elev: 18

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.0	6.5	-3.9	4.5	1.0	2.1	6.1	7.7	36.1	33.9	32.4	33.7	9.7	60	0.721

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	77.1	67.0	73.8	66.3	71.3	64.4	69.6	74.8	67.5	72.5	65.4	70.3	11.5	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	101.3	73.1	65.5	94.3	70.8	63.1	86.6	68.6	33.6	74.7	31.9	72.5	30.3	70.2	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
29.7	26.6	23.7	-1.7	81.7	8.9	4.5	-8.1	84.9	-13.3	87.6	-18.4	90.1	-24.9	93.4	
			DB												
			WB	-2.2	69.9	8.9	3.0	-8.6	72.1	-13.8	73.8	-18.8	75.5	-25.3	77.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.5	25.8	26.3	30.9	38.8	50.4	58.2	64.8	63.6	56.5	45.7	39.1	32.4
	DBStd	15.14	8.83	9.96	6.67	4.98	7.06	5.05	5.11	3.98	4.68	5.58	6.76	8.38
	HDD50	3467	752	663	591	339	83	2	0	0	9	154	328	546
	HDD65	7594	1217	1083	1056	787	455	212	68	72	256	599	778	1011
	CDD50	1451	0	0	0	2	94	249	460	423	204	19	0	0
	CDD65	104	0	0	0	0	1	9	63	30	1	0	0	0
	CDH74	269	0	0	0	0	3	23	186	57	0	0	0	0
	CDH80	29	0	0	0	0	0	0	22	7	0	0	0	0
	WSAvg	12.2	12.1	11.6	12.0	10.9	10.9	11.9	11.0	11.7	12.4	13.5	13.7	14.1
	PrecAvg	25.9	2.0	1.3	1.5	1.5	1.2	1.9	2.2	3.4	2.3	3.0	2.8	2.4
Precipitation	PrecMax	33.2	3.9	3.5	3.6	2.2	2.5	3.0	5.1	5.9	4.5	6.9	4.5	5.5
	PrecMin	19.2	0.1	0.0	0.3	0.2	0.2	0.4	0.0	1.2	1.0	0.4	0.4	0.6
	PrecStd	3.8	1.1	1.0	0.7	0.6	0.7	0.7	1.5	1.3	1.0	1.7	1.1	1.2
	PrecStd	3.8	1.1	1.0	0.7	0.6	0.7	0.7	1.5	1.3	1.0	1.7	1.1	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.9	41.0	46.7	58.0	73.1	77.2	83.0	80.5	69.8	58.0	51.2	47.5
		MCWB	38.9	36.5	39.2	47.9	60.0	64.2	68.9	68.6	62.2	53.4	49.4	45.4
	2%	DB	39.2	39.0	42.8	52.9	68.0	72.4	78.6	74.6	66.1	55.8	49.9	45.0
		MCWB	37.5	36.7	37.9	44.3	57.1	63.2	67.6	66.3	60.5	53.2	48.7	42.9
	5%	DB	37.5	37.4	40.4	49.0	64.9	68.6	75.6	71.9	64.1	54.5	48.2	43.2
		MCWB	36.0	35.9	36.7	42.7	56.1	60.4	68.0	64.6	59.2	52.0	46.8	41.2
	10%	DB	35.9	35.9	38.6	45.6	61.6	66.0	72.9	69.5	62.7	53.1	46.7	41.8
		MCWB	34.7	34.8	35.8	40.7	54.7	58.7	66.5	63.0	58.6	50.6	45.0	39.9
	0.4%	WB	39.6	37.8	40.9	48.7	61.6	66.8	71.9	71.3	63.5	55.1	49.9	45.9
		MCDB	40.5	39.2	44.1	56.5	68.6	74.0	77.6	76.1	67.3	56.2	50.3	46.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.9	36.9	38.6	45.2	59.5	64.0	70.4	68.4	61.6	53.7	48.7	43.4
		MCDB	38.9	38.1	41.3	51.1	66.1	70.2	75.8	73.4	64.9	55.3	49.5	44.6
	5%	WB	36.3	36.0	37.3	43.2	57.5	62.1	68.9	66.2	60.3	52.6	47.0	41.4
		MCDB	37.3	37.2	39.8	47.6	63.4	67.7	74.1	70.5	63.3	54.2	48.1	42.9
	10%	WB	34.9	34.9	36.0	41.7	54.9	59.9	67.1	64.3	59.2	51.0	45.1	40.0
		MCDB	35.8	36.0	38.0	45.2	60.8	64.7	72.1	68.4	62.0	52.7	46.3	41.5
Mean Daily Temperature Range	5% DB	MDBR	7.3	7.4	9.1	9.9	10.3	8.9	8.6	7.4	6.9	6.1	5.0	5.9
		MCDBR	5.2	6.5	9.1	15.4	16.2	14.1	12.4	10.4	8.7	5.6	5.3	6.4
		MCWBR	4.9	5.3	6.3	8.7	8.0	7.2	5.7	5.9	5.2	5.0	5.4	6.0
	5% WB	MCDBR	5.1	6.0	8.3	13.3	15.1	12.4	10.3	7.6	7.2	5.2	5.2	6.2
		MCWBR	5.2	5.2	6.3	8.0	8.1	7.4	5.6	5.6	5.0	5.1	5.3	6.0
Clear-Sky Solar Irradiance	taub		0.278	0.314	0.334	0.373	0.369	0.361	0.389	0.380	0.351	0.334	0.304	0.386
	taud		2.156	2.237	2.310	2.317	2.404	2.457	2.401	2.421	2.488	2.397	2.211	2.608
	Ebn at Noon		149	213	251	261	272	276	264	257	246	207	137	103
	Edh at Noon		13	22	28	33	33	32	33	30	24	19	13	9
	RadAvg		75	260	743	1248	1708	1829	1712	1340	820	353	97	46
All-Sky Solar Radiation	RadStd		12	49	88	122	158	135	162	131	99	55	19	6

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)		+0.74	N/A	N/A	N/A	N/A	N/A	-239	N/A	N/A

Nomenclature: See separate page